M31995 & M32274-DL Data Analysis and Visualisation Coursework 2 (2024-2025)

This is an **individual** assignment and carries 50% of the module mark. It covers the following learning outcomes for the module:

- **Be able to apply common algorithms/models and analyse the choice of performance metrics**
- **Demonstrate the ability to use analytical packages for quantitative and visual analysis in different business contexts**
- **Visualise data effectively, and understand the key principles of data communication**

There are two sections in this coursework. Under Section A, you are expected to answer all the questions, which account for 70% of this coursework (therefore 35% of your final module mark). Under Section B, you are expected to answer one discussion question, which accounts for 30% of this coursework (therefore 15% of your final module mark). The marks for Presentation and English language are embedded in the two Sections.

Sections A and B of this coursework assignment can be found on **Page 5** and **7**, respectively.

Submission Date: **Tuesday 20th May 2025 by 13:00 (UK time).**

Submission Hand in Arrangements: Students are required to submit their coursework assignment to **Turnitin** via the link on the module Moodle.

There are mark penalties for late hand in. (Corruption of computer files is not an adequate excuse for late hand in, as work should be adequately backed up.)

If there is a problem with Moodle at the time of submission students should email their coursework to submissions@port.ac.uk with the following details:

Student Number

Module Name / Module number

Course name that you are studying

Screenshot of the error

PLEASE NOTE:

This assessment is eligible for the automatic 48-hour extension. The end of the Extension Period will be 13.00 on Thursday 22nd May 2025.

If you're facing short-term circumstances that mean you're not able to submit your assessment by the published deadline, you can make use of the Automatic Extension. This gives you a further 48 hours to submit your work - and you don't have to do

anything to request the extension. The end of the Automatic Extension period will be 13.00 on Thursday 22nd May 2025.

Please note, you won't have access to any academic support relating to completing your assessment during the 48-hour extension period.

Up until the point of the original deadline, you may overwrite your submission. Once the original deadline has passed, you may not resubmit your work even if it is within the automatic 48-hour extension period. If you need to make use of the automatic 48-hour extension, you may only submit your work once during the 48-hour period.

Word Limit: For Sections A and B combined: 1600 words excluding cover page, footnotes, bibliography, appendices, tables/charts/graphs, but including all in-text citations. (There will be a suggested word limit for each Section but you can take it as a loose guidance. As long as the total word count does not exceed 1,600, it is fine.) An abstract is NOT required. The word count should be stated at the top of the assignment. A falsely stated word-count is an assessment offence which may result in a penalty, including the reduction of the mark to 0%. Note, that footnotes should be used to reference sources only. Examiners are free to disregard footnotes that contain inappropriate information or information that should belong in the main text. Coursework that is over the stated word limit will result in a penalty of 10% of the original mark awarded. **For the avoidance of doubt, the penalty will be applied to any work that exceeds the stated word limit of 1600 words excluding cover page, bibliography and appendix.**

Referencing requirements: Students must reference all sources using the APA 7th Edition. Guidance on this method of referencing can be found at https://library.port.ac.uk/ref/page44.html. Reference should be made to the primary source, except when the primary source can no longer be obtained. Poor citation of sources will result in a loss of marks.

Referencing is required to give intellectual credit to your source, help readers recover your source easily and to avoid being accused of plagiarism.

Students are reminded that the University will not tolerate academic dishonesty in any form. This is cheating.

Plagiarism: Students are reminded of the need to avoid plagiarism. The University Regulations describe plagiarism as:

> the incorporation by a student in work for assessment of material which is not their own, in the sense that all or a substantial part of the work has been copied without any adequate attempt at attribution, or has been incorporated as if it were the student's own when in fact it is wholly or substantially the work of another person or persons.

Any student suspected of plagiarising will be referred for an Academic Misconduct Hearing.

Students should ensure that all sources are fully cited and in the reference list and that indentation or quotation marks (as appropriate) are used when quoting. Failure to include a reference list (where needed) will result in a 5% penalty, unless the lecturer/tutor has advised you that a reference list is not required.

Formatting: The work should be word processed. Font size should be between 11 and 14 and 'easy to read' e.g., Calibri, Arial, Times New Roman. Line spacing should be between 1.5 and 2 with (approx.) 2.5 - 4 cm margins all round. The Header must include the student number and the Footer must include a page number.

Electronic Copy of Work: Students should retain an electronic copy of their coursework, so that it may be checked by a member of staff should a member of staff feel the need to do so. Tutors are entitled to request an electronic copy of coursework if they have any doubt about the accuracy of the stated word count and/or any suspicion of plagiarism. Failure to send an electronic copy of the coursework to a member of staff who has asked for a copy may result in a penalty.

If any student has a query about any of the above matters and wishes to obtain clarification or further information, please contact your Module Co-ordinator or personal tutor.

If any student requests IT support for the submission, please contact the university IT service desk: +44 (0)23 9284 7777.

Marking Criteria: Marking will be done in accordance with the grade descriptors in the table below and the categorical marking criteria for the University of Portsmouth for PG level 7 as set out in Table 6 of the Assessment Regulations for Undergraduate and Postgraduate. Marks and feedback will be available within 20 working days after the assessment deadline.

	Level 7
80+	As below plus: Potentially publishable work
70-79	Excellent work - able to express an original reasoned argument in a lucid manner by reviewing & critiquing a wide range of material. Original, critical thinking based on outstanding insight, knowledge & understanding of material. Material contributes to current understanding & is of potentially publishable quality in terms of presentation and content ▪ Wide reaching research showing breadth & depth of sources
60-69	As below plus: ▪ Clear, balanced coherent critical & rigorous analysis of the subject matter. Detailed understanding of knowledge & theory expressed with clarity ▪ Extensive use of relevant & current literature to view topic in perspective, analyse context & develop new explanations and theories
50-59	As below plus: ▪ Detailed review and grasp of pertinent issues & a critical contextual overview of the literature. Thorough knowledge of theory and methods & uses this to underpin arguments and conclusions ▪ Confidence in understanding and using literature
40-49	▪ Demonstrates grasp of key concepts & an ability to develop & support an argument in a predominately descriptive way with valid conclusions drawn from the research ▪ Familiarity with key literature which is cited and presented according to convention ▪ Logical & clear structure, well organised with good use of language and supporting material
30-39 **0-29**	FAIL Some knowledge of relevant concepts & literature but significant gaps in understanding and/or knowledge. Little attempt at evaluation, conclusions vague, ambiguous & not based on researched material. Limited or inappropriate research. Deficits in length, structure, presentation &/or prose

SECTION A (70%):

You will use the data provided to answer the following questions. The data are available in the Excel file attached.

The word count for Questions 1 and 2 in total is not expected to be more than 1,200 words. If it exceeds 1,200, you may have spent the words on unnecessary discussions. It is fine to have fewer words, as long as it is enough to explain your answers. You can attach screenshots from Excel or Stata as evidence and the screenshots do not count towards the word limit.

References can be used to support your explanation but this is not compulsory.

Answers to each question should be clearly labelled.

Question 1

In your role as a Data Analysis Specialist at a leading automotive market research firm, you have been approached by a renowned automobile manufacturer seeking to gain deeper insights into the pricing strategies for their vehicle lineup. The manufacturer is particularly interested in understanding how various factors, including the sale date of a vehicle model, its age, proximity to urban centres, and the number of dealerships nearby, influence its market price.

The data for Task 1 is presented on the sheet **'Automobile'** of the excel file '**Data_CW2.xlsx'**

Question 1.1 (a) Build a linear regression model to predict vehicle value. (b) Interpret your regression results for your customer. (14 marks)

Question 1.2 Your group manager questions whether the model may suffer from potential heteroskedasticity and multicollinearity issues, but you do not agree. (a) please briefly explain the concepts of heteroskedasticity and multicollinearity; (b) and show evidence on why your results in Question 1 do not suffer from such issues; (c) what are possible methods to address the issues of heteroskedasticity and multicollinearity? (12 marks)

Question 1.3 One of your colleagues suggests using a nonlinear model instead of a linear model. Please analyse the data and evaluate whether a nonlinear model is a better choice. (5 marks)

31 marks in total

Question 2

You are building an algorithm to determine varieties of iris (versicolor or virginica). You have collected data for a sample of iris. The data include sepal length, sepal width, petal length, petal width, and the variety of the flower. The data are available in the Excel file attached.

The data for Task 2 is presented on the sheet **'Iris'** of the excel file '**Data_CW2.xlsx'**

Question 2.1 Use a Kth-Nearest Neighbour model and the Euclidean distance algorithm to determine the variety of the following observation (6.6, 3.2, 5.1, 1.5). Use relevant theories of the model to justify your choice. (10 marks)

Question 2.2 When we use the K-NN model to classify an unseen instance, a) what might be the problem if we set the value of K to an even number? Please provide the evidence combining the results from Question 2.1. b) What could be a possible solution? (6 marks)

Question 2.3 Apply 5-NN, 7-NN, and 9-NN models, a) show the confusion matrix, and b) calculate the accuracy of these models. c) Which model do you think is the best? Please explain your choice. (16 marks)

32 marks in total

7 marks in total for the presentation and language of Section A

SECTION B (30%):

You have studied several aspects of effective charts including the following:

- Answer most of your questions.
- Help compare and rank.
- Show the correlation between different measures or variables.
- Emphasize the most important data.
- They are not overloading views.
- Show a limited number of colours and shapes in a single view.

Besides, there are several questions to ask before producing effective charts such as:

- What are you trying to say? What is the story that you want to tell?
- Who is your audience, their experience to understand the graphs?
- What questions might your audience have?
- What answers are they searching for?
- What is an appropriate chart type to meet your audience's needs?
- Can your audience understand the visualisation in 30 seconds or less, without additional information?

Requirements:

Choose a chart or dashboard that is relevant to the business world, such as KPIs charts or financial charts, produced and published by major companies like Amazon, IBM, etc. Ensure that your selection is informative and displays a professional standard.

Write a report of no more than 400 words to critically evaluate the effectiveness of the chart/dashboard you have chosen. Your evaluation should consider the principles of effective charts and address the questions above.

30 marks in total, including 3 marks for presentation and English